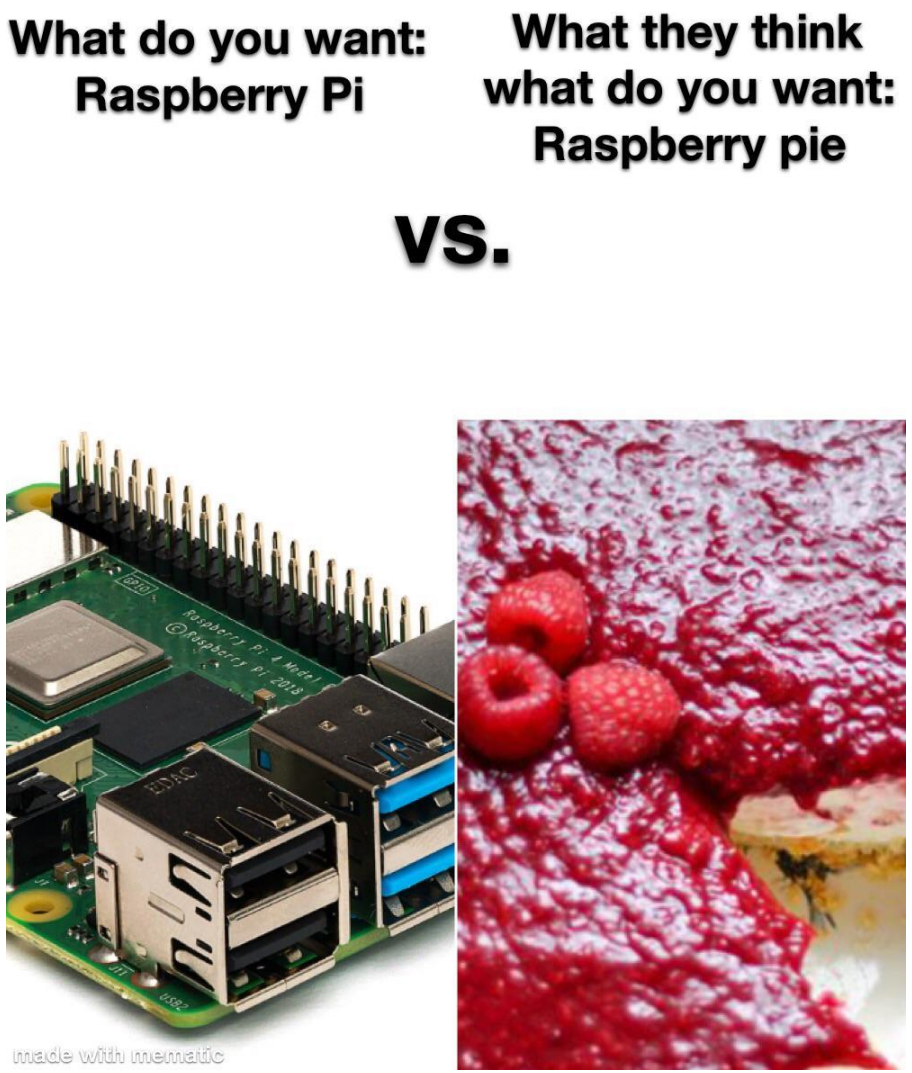
Modul 321 Projekt

Leonardo Evic

Inhaltsverzeichnis

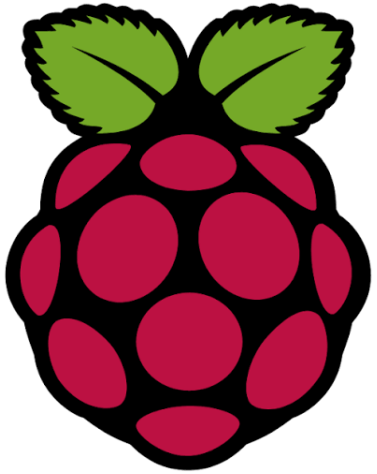
- Einführung
- Technische Aspekte
- Demo
- Lernaspekte



Einführung

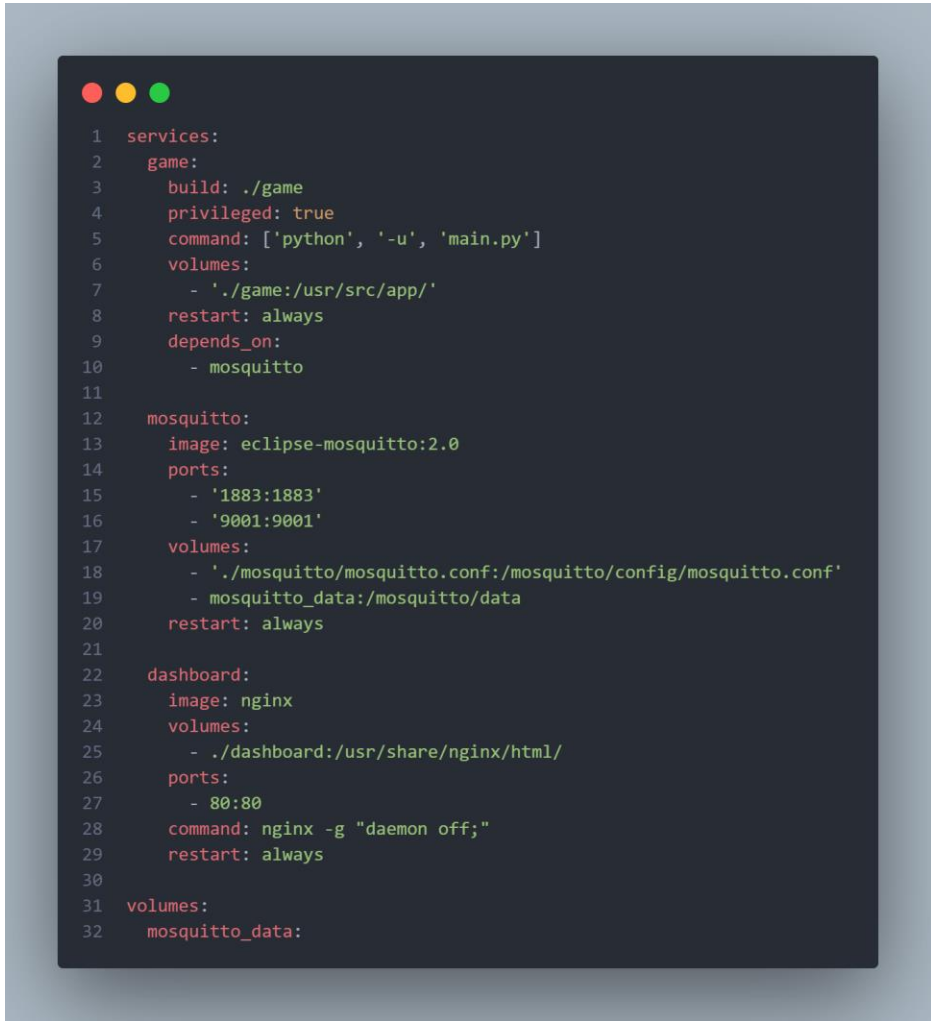
- Zombie-Spiel in Python
- Verwendet Joy-Pi Komponenten
  - Raspberry Pi
  - Buttons
  - Touch-Sensor
  - LED-Matrix





Technische Aspekte - Aufbau

- Docker Compose
  - Spiel
  - Mosquitto MQTT
  - Dashboard



```yaml
services:
  game:
    build: ./game
    privileged: true
    command: ['python', '-u', 'main.py']
    volumes:
      - './game:/usr/src/app/'
    restart: always
    depends_on:
      - mosquitto

  mosquitto:
    image: eclipse-mosquitto:2.0
    ports:
      - '1883:1883'
      - '9001:9001'
    volumes:
      - './mosquitto/mosquitto.conf:/mosquitto/config/mosquitto.conf'
      - mosquitto_data:/mosquitto/data
    restart: always

  dashboard:
    image: nginx
    volumes:
      - ./dashboard:/usr/share/nginx/html/
    ports:
      - 80:80
    command: nginx -g "daemon off;"
    restart: always

volumes:
  mosquitto_data:
```

Technische Aspekte - Spiel

- main.py – Einstiegspunkt
- game.py – Hauptklasse des Spiels
- controls.py – Steuerung
- screen.py – LED-Matrix



Technische Aspekte - Spiel

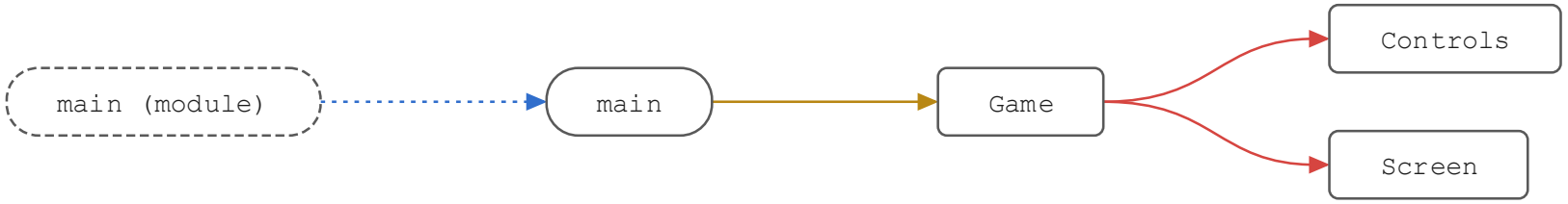
Python Class Map

game - 2026-10-25 - showing: creates (member), creates, holds, calls

Legend - relationship kinds

creates (member)	self.attr = Foo() - the instance is stored on the object (owned for its lifetime)
creates	Foo() on the fly in a method or function body - not stored on self
holds	typed __init__ param or class field - holds an instance it did not create
calls	calls a project function that (transitively) creates instances
inherits	subclassing (is-a) - drawn with a hollow triangle arrowhead (hidden in this view)

Nodes: square corners = class, pill = function, dashed border = module, dotted = external



Technische Aspekte - Dashboard

- HTML/CSS/JS Webseite
- Zeigt den Spielstand an
- Zeigt die Punktezahl an
- Datenaustausch mit MQTT
  - MQTT.js

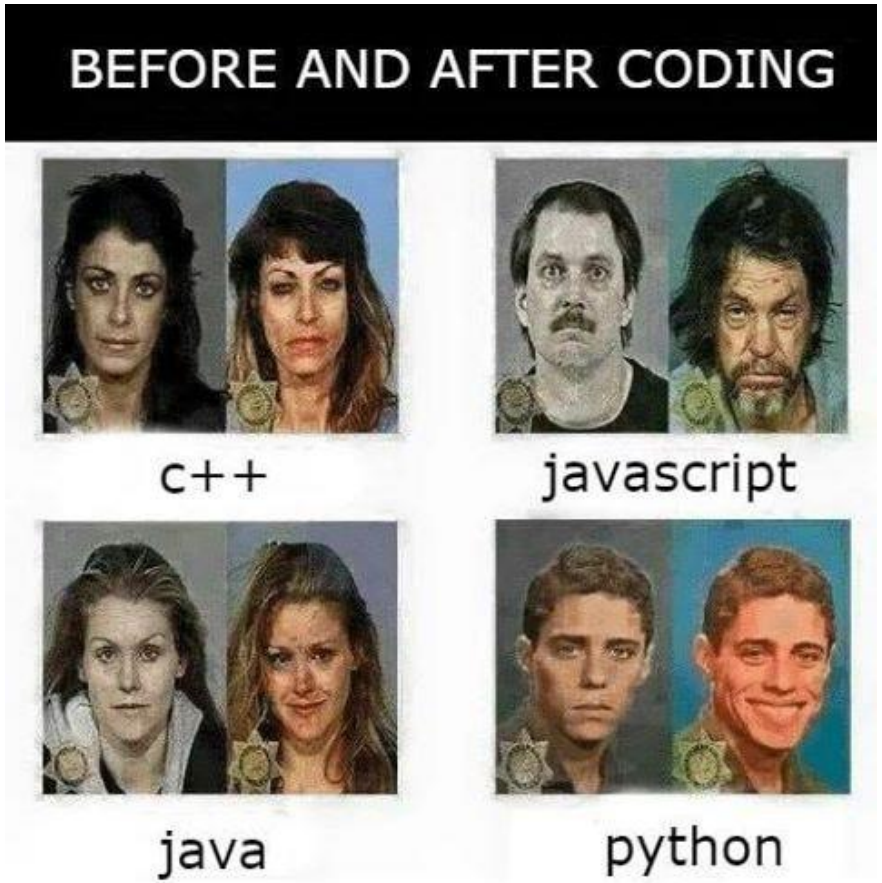


Technische Aspekte –MQTT Broker

- Mosquitto MQTT Broker
- Spiel (game.py) veröffentlicht Daten
- Dashboard abonniert Topics
- Retained Messages
  - Daten bei Refresh anzeigen



Demo



Lernaspekte

- Python + MQTT Refresher
- Abfragen von Sensoren / Buttons
- Fehlerbehandlung
  - Race Conditions mit threading.Lock
  - Docker Cache

Fragen



Ende.